

Edge-Cloud Collaborative Computing: A Survey

Distributed Intelligence and Model Optimization

Your Name

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Overview

- Introduction and Motivation
- Edge-Cloud Computing Paradigm
- Model Optimization Techniques
- Resource Management Strategies
- Privacy and Security Mechanisms
- Practical Applications
- Future Research Directions
- Conclusion

Introduction and Motivation

- Rise of IoT and real-time applications demands low latency and efficient processing.
- Edge-Cloud Collaborative Computing (ECCC) integrates edge and cloud resources.
- Challenges: resource constraints, model deployment, and security concerns.
- Applications: autonomous driving, healthcare, industrial automation.

Edge-Cloud Computing Paradigm

- Three-tier architecture: cloud, edge, end devices.
- Reduces latency and improves scalability by distributing computation/storage.
- Distributed intelligence via task, data, and model parallelism.
- Optimizes performance, cost, and latency metrics.

Model Optimization Techniques

- **Model Compression:**
 - Pruning (e.g., Hybrid SD, PSFed).
 - Quantization (e.g., 8-bit integer representations).
 - Knowledge distillation.
- **Adaptation:** Neural Architecture Search (NAS) tailors models to dynamic environments.
- Collaborative strategies enhance model deployment across distributed systems.

Resource Management Strategies

- AI-driven techniques: Deep Reinforcement Learning (DRL), multiobjective optimization.
- Dynamic resource allocation and task offloading to minimize latency/energy.
- Advanced networking: SDN, NFV, network slicing, 6G.
- Ensures efficient data flows and Quality of Service (QoS) guarantees.

Privacy and Security Mechanisms

- Federated learning enables collaborative model training while protecting data.
- Privacy-preserving techniques address data security in distributed systems.
- Robust security mechanisms safeguard against vulnerabilities in ECCC.

Practical Applications

- **Autonomous Driving:** Real-time decision-making with low latency.
- **Healthcare:** Efficient processing for patient monitoring and diagnostics.
- **Industrial IoT:** Supports mission-critical automation tasks.
- Performance analysis and benchmarking evaluate ECCC deployments.

Future Research Directions

- Deploying Large Language Models (LLMs) at the edge.
- Integration with 6G networks for enhanced connectivity.
- Exploring neuromorphic and quantum computing paradigms.
- Addressing heterogeneity, real-time processing, and scalability challenges.

Conclusion

- ECCC enables efficient, low-latency AI applications in distributed networks.
- Integrates model optimization, resource management, and advanced networking.
- First survey to target distributed intelligence and model optimization in ECCC.
- Highlights research gaps and positions ECCC as a transformative paradigm.

Source: Liu et al., IEEE Communications Surveys & Tutorials, 2025